

Notes on Multivariable Calculus

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1 Geometric Intuition and Motivation

What are we dealing with:

- $f(x, y) = x^2 + y$, $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ [functional]
- $f(x, y) = \begin{bmatrix} 3x \\ 2y \end{bmatrix}$, $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ [vector-valued functions]
- $f(x, y) = \begin{bmatrix} x^2 \\ y^2 \\ 2x - y \end{bmatrix}$, $f : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ [parametric surfaces]

Different ways to visualize:

- functionals can be represented as graphs with one of the axes (typically the z -axis) serving for the function value itself, or contour plots
- vector-valued functions $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ can be visualized as a vector field.
- vector-valued functions $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ can also be visualized by watching the entire input space in $\mathbb{R}^2$ move to the output space in $\mathbb{R}^2$.
- parametric surfaces for visualizing a mapping of a plane (say, $\mathbb{R}^2$) move into a surface embedded in $\mathbb{R}^3$.

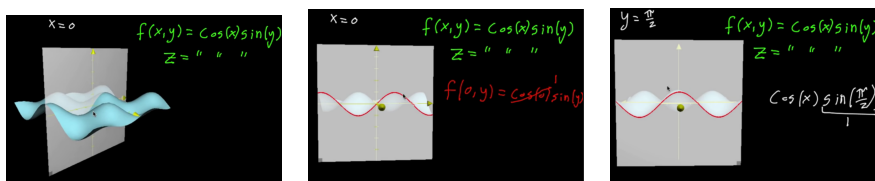


Figure 1: Slicing Graphs

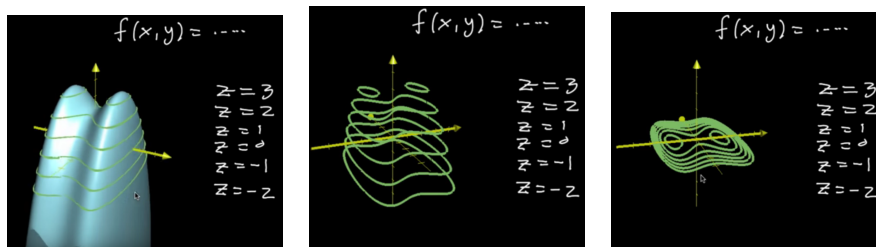


Figure 2: Contour Plots

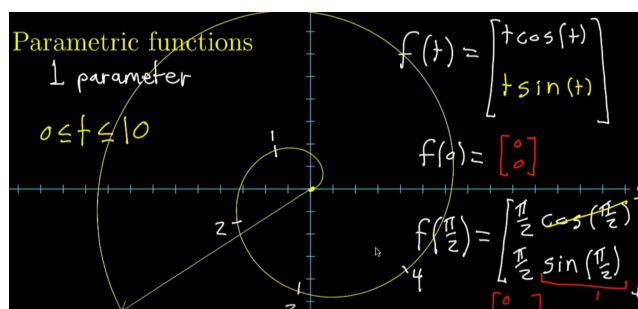


Figure 3: Parametric surface

1.0.1 Visualizing Functionals: Graphs and Contour plots

For 3-dimensional graphs, slicing the graph for a fixed value of the input can give us some insight. For example, in the graph of $f(x, y) = \cos(x) \sin(y)$ (see fig 1), we can set $x = 0$ or $y = \frac{\pi}{2}$ and examine the behavior.

If we slice the graph at z for some constant value. That would give rise to the contour plot of the graph (see fig 2).

1.0.2 Visualizing parametric surfaces

Parameter here is a fancy name for the input, e.g. $f(t) = \begin{bmatrix} t \cos(t) \\ t \sin(t) \end{bmatrix}$.

Best way to visualizing these objects is to look at just the output space of the function.